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Chapter 6

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Abstract

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Chapter 6
Photodetectors

Andreas Beling and Joe C. Campbell

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6.1 Fundamentals

The light absorption process in a semiconductor photodetector for photogeneration of electron-hole-pairs is based on the internal photoelectric effect and requires the photon energy $h\nu$ to be at least equal to the bandgap energy E_g of the absorber material. Only then is the available energy of one photon sufficient to excite an electron from the valence band to the conduction band leaving a hole in the valence band. For this band-to-band transition, the upper wavelength limit for photon absorption is given by:

$$\lambda_g\ [\mu\mathrm{m}] = \frac{1.24}{E_g\ [\mathrm{eV}]} \tag{6.1}$$

Under the influence of an electric field, that is established by an applied bias voltage, electrons and holes are swept across the absorber which results in a flow of photocurrent in the external circuit [1]. The external quantum efficiency η_{ext} quantifies the ability of the photodiode (PD) to transform light into an electrical current

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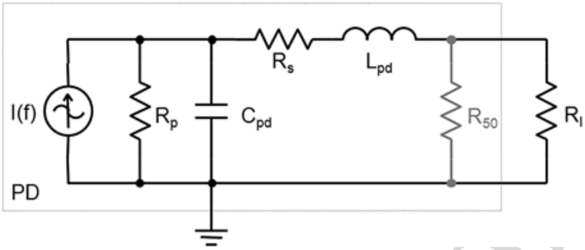

Fig. 6.1 Equivalent circuit of a lumped element photodetector

and is defined as the number of charge carrier pairs generated per incident photon:

$$\eta_{ext} = \frac{I_{pd}}{q} \cdot \frac{h\nu}{P_{opt}} \tag{6.2}$$

where I_{pd} is the photogenerated current by the absorption of the optical input power P_{opt} at frequency ν, and q is the elementary charge (-1.602×10^{-19} C). Ideally $\eta_{ext} = 1$, that is each photon generates one electron hole pair. However, it will be shown below that in practice photodiodes usually exhibit $\eta_{ext} < 1$ because of several effects including finite absorber thickness, carrier recombination, optical reflections and coupling losses. A common figure of merit is the responsivity R_{pd}, defined as the ratio of photocurrent to optical input power. Based on (6.2) we write:

$$R_{pd} = \frac{I_{pd}}{P_{opt}} = \frac{\eta_{ext}\lambda\ [\mu\text{m}]}{1.24} \frac{A}{W} \tag{6.3}$$

Using (6.3), $\eta_{ext} = 1$, and a wavelength of $\lambda = 1.55$ µm we find that the maximum achievable responsivity is $R_{pd} = R_{ideal} = 1.25$ A/W. Whenever R_{pd} depends on the state of polarization of the incoming light, the definition of the polarization dependent loss is useful:

$$PDL = 10 \log\left(\frac{R_{\max}}{R_{\min}}\right)\ [\text{dB}] \tag{6.4}$$

Here, $R_{\max}$ and $R_{\min}$ are the maximum and minimum responsivities for all states of polarization. In telecommunication systems photodiodes are required to detect optical signals modulated at high data rates. Thus, another important figure of merit is the opto-electrical 3 dB bandwidth, which is defined as the frequency range from DC to the cut-off frequency $f_{3\,\text{dB}}$. The latter is the frequency at which the electrical output power has dropped by 3 dB below the power value at very low frequency. The RC-time constant and the carrier transit times are the two crucial bandwidth limitations of a photodiode. Considering the RC-effect, Fig. 6.1 shows a simple equivalent circuit of a lumped element photodetector for which the device length is much shorter than the electrical signal wavelength. The photodiode is described as an ideal current source $I(f)$ in parallel with the junction capacitance, C_{pd}, and resistance, R_p, and a series resistance, R_s. R_l is the external load resistor and is usually 50 Ω. In order to achieve a matched transition to the load impedance, R_l, an optional termination resistor, R_{50} ($= 50$ Ω), may be implemented close to the photodiode. Thus, the effective load is reduced to 25 Ω, but at the expense of losing half the output current. In most cases the parallel leakage current is small compared to the photocurrent and therefore R_p becomes very high and can be omitted. The inductance L_{pd} may originate from electrical interconnections or air bridges. It is usually in the pH-range and can be neglected in photodetectors with bandwidths up to 20 GHz. With these simplifications the RC-limited 3 dB cut-off frequency is calculated from the equivalent circuit as

$$f_{RC} = \frac{1}{2\pi R_{eff} C_{pd}} \tag{6.5}$$

with $R_{eff} = R_s + R_l R_{50}/(R_l + R_{50})$.

The second bandwidth constraint is the carrier transit time, which is the time a photogenerated electron or hole takes to travel through the active region prior to being collected by the contacts. Assuming uniform photogeneration in an absorber with thickness d_{abs}, the transit time-limited bandwidth can be well estimated using [2]:

$$f_t \approx \frac{3.5\bar{v}}{2\pi d_{abs}} \tag{6.6}$$

where $\bar{v}$ is the averaged carrier velocity. The resulting 3 dB bandwidth is given by the expression:

$$f_{3\,\mathrm{dB}} \approx \sqrt{\frac{1}{\frac{1}{f_{RC}^2} + \frac{1}{f_t^2}}} \tag{6.7}$$

Taking the product of the bandwidth (6.7) and quantum efficiency (6.2) one can define the bandwidth-efficiency product in units of [GHz]. As will be shown in the following paragraphs, the bandwidth-efficiency product of photodiodes is generally limited.

In the absence of light the photodiode's dark or leakage current dominates. Generally several effects contribute to the dark current with their actual magnitudes depending on the bias voltage, material properties, and detector design. The current that is common to all junction diodes is the diffusion current. However, in most photodiodes the diffusion current is considerably smaller than the generation current [3, 4], which originates in the depleted absorption layer from impurities within the bandgap and can be reduced by improving material quality. Additionally, tunneling and impact ionization currents can be observed once the electric field exceeds 100 kV/cm. While the tunneling current is generally undesirable, impact ionization can be used to provide an internal gain mechanism to amplify the photocurrent. Photodiodes that exploit impact ionization effects are referred to as avalanche photodiodes (APD) and will be discussed in Sect. 6.2.3.

Photocurrent and dark current generate shot noise, which, together with thermal noise, degrades the sensitivity of the photodiode. Shot noise accompanies any generated current within a photodetector and is related to the statistical nature of the carrier transport and the photon detection process. The mean square noise current of shot noise for a photodiode is given by

$$\langle i^2_{shot} \rangle = 2q(I_{pd} + I_d)\Delta f \tag{6.8}$$

I_{pd} is the photocurrent, I_d the dark current and Δf is the bandwidth.

The thermal noise or Johnson noise is due the thermal motion of electrons in conductors and is generated in all resistances of the photodiode. The mean square thermal noise current of a resistor R for a given bandwidth Δf is given by the expression

$$\langle i^2_{th} \rangle = \frac{4kT\Delta f}{R} \tag{6.9}$$

where k is Boltzmann's constant and T is the temperature. It is worth noting that $\langle i^2_{th} \rangle$ does not depend on the photocurrent and dark current whereas $\langle i^2_{shot} \rangle$ does. Since shot noise and thermal noise are uncorrelated, the total noise is found from adding individual noise contributions as sums of squares. Once the total noise in a photodiode is known, the signal-to-noise ratio, SNR, can be determined. The electrical SNR is defined as the ratio of signal power to noise power and can be computed for a given average photocurrent, I_{pd}, using the relation

$$SNR = \frac{I^2_{pd}}{\langle i^2_{shot} \rangle + \langle i^2_{th} \rangle} \tag{6.10}$$

which assumes that the power varies as the square of the current.

An important measure in digital communication systems is the bit-error rate (BER) given by the probability of false identification of a bit by the decision circuit in the receiver. The decision circuit compares the sampled signal to a reference value, the decision threshold. If the signal is greater than the decision threshold, it indicates that a "1" was detected, otherwise a "0". As in binary systems there are only two possible signal levels, that for a "1", I_1, and that for a "0", I_0. Each of these signal levels may have a different average noise associated with it. In order to calculate the overall probability of a bit error, the SNRs of both signal levels have to be taken into account which leads to the definition of the Q-factor:

$$Q = \frac{I_1 - I_0}{\sqrt{\langle i^2_1 \rangle} + \sqrt{\langle i^2_0 \rangle}} \tag{6.11}$$

In this expression I_1, I_0, and $\langle i^2_1 \rangle$, $\langle i^2_0 \rangle$ are the photocurrents and the Gaussian mean square noise currents associated with the received bits "1" and "0", respectively. Assuming a constant but optimum threshold level in the decision circuit, the BER is related to the Q-factor by the following equation:

$$BER = \frac{1}{2}\mathrm{erfc}\left(\frac{Q}{\sqrt{2}}\right) BER = \frac{1}{2}\mathrm{erfc}\left(\frac{Q}{\sqrt{2}}\right) \approx \frac{\exp(-Q^2/2)}{Q\sqrt{2\pi}} \tag{6.12}$$

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where erfc represents the complementary error function. The approximate form of (6.12) is accurate for $Q > 3$. A bit-error rate of 10^{-12} or a Q-factor of 7 corresponds to a probability of $1 : 10^{12}$ that a bit is identified incorrectly and is commonly called “error-free” reception.

The receiver sensitivity is a common figure in optical communication systems. It is the minimum received average optical power P_{rec} that is necessary to achieve error-free detection at a given bitrate. Considering on-off modulated signals with optical power P_1 and P_0 in bits “1” and “0”, respectively, the average power is $0.5(P_1 + P_0)$. Here we consider a more general case, that is $P_0 \neq 0$. The fact that some optical power is received during “0” bits is common to most fiber links and is quantified by the extinction ratio, $r_e = P_0/P_1$. Using Eqs. (6.3) and (6.11), the receiver sensitivity can be approximated as:

$$P_{rec} = \left(\frac{1+r_e}{1-r_e}\right)\frac{Q}{2R}\left(\sqrt{\langle i_1^2\rangle} + \sqrt{\langle i_0^2\rangle}\right) \approx \left(\frac{1+r_e}{1-r_e}\right)\frac{Q}{R}\sqrt{\langle i_{th}^2\rangle} \tag{6.13}$$

where shot noise from I_{pd} and I_d has been neglected.

It should be noted, that the sensitivity is usually used to characterize the entire optical front-end including photodiode and electrical amplifiers. This requires modification of (6.13) in order to include noise contributions from the amplifier. Similarly, if the receiver contains an optical pre-amplifier, the amplifier’s noise figure has to be taken into account. In this case P_{rec} is typically referred to the optical power before amplification.

The ultimate detection limit of an ideal receiver (i.e. no thermal noise, no dark current, 100 % quantum efficiency, and $r_e = 0$) is given by the quantum limit. Assuming an ideal receiver, the sensitivity for a BER of 10^{-12} is given by [5]:

$$P^*_{\min} \approx 13.5 \cdot h\nu \cdot B \tag{6.14}$$

where $h\nu$ is the photon energy, B is the bitrate and 13.5 is the average number of photons per bit. For example, at a bitrate of 40 Gbit/s and an optical signal wavelength of 1.55 μm, the quantum limited sensitivity $P^*_{\min}$ is −42 dBm.

6.2 Photodiode Types

6.2.1 p–i–n Photodiode

In order to achieve a high optoelectronic conversion efficiency and high bandwidth the p–i–n photodiode is widely used. The p–i–n PD consists of an intrinsic absorber, sandwiched between highly doped n^+- and p^+-layers which give rise to a space charge region. Compared to a simple p–n-junction, this design allows for a lower junction capacitance and provides an additional degree of freedom in designing the thickness of the depleted high-field region. In contrast to a homojunction PD where

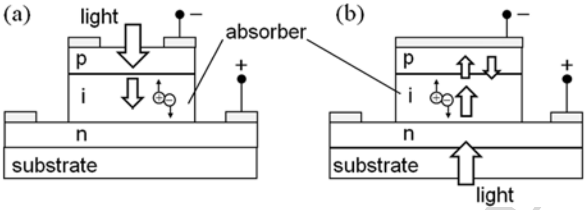


Fig. 6.2 Top (**a**) and back (**b**) illuminated p–i–n photodiode

the p-, i-, and n-layers have the same bandgap, a heterostructure p–i–n PD potentially provides higher speed, as slow diffusion photocurrents arising from carriers generated in undepleted material are suppressed. In this design the contact layers exhibit bandgap energies higher than the photon energy and are consequently transparent at the operation wavelength. In order to avoid slow carrier trapping effects at the heterojunction interfaces grading of the bandgap can be employed to smooth the discontinuities in the band structure [6].

A schematic of a top-illuminated p–i–n photodiode is shown in Fig. 6.2(a); this simple structure has become a standard commercial product. For high-speed operation the device is reverse biased by an externally applied voltage to completely deplete the absorber and create an electric field to facilitate carrier transport. Once an electron-hole-pair is created, the carriers drift at their saturation velocities in opposite directions toward the electrodes and contribute to the photocurrent. The device responsivity can be written as

$$R_{pd} = R_{ideal}(1 - R_0)\left(1 - e^{-\alpha d_{abs}}\right) \tag{6.15}$$

where α is the absorption coefficient, d_{abs} is the absorber thickness and R_0 is the reflectance at the air-semiconductor interface. Assuming negligible carrier trapping at energy band discontinuities the bandwidth is well approximated by (6.7).

It is well known that for this device structure there is a performance trade-off between quantum efficiency and bandwidth associated with the thickness of the absorber region. As thicker absorption layer provides higher quantum efficiency this also results in longer carrier transit time which reduces the bandwidth. A second bandwidth constraint is due to the photodiode RC time constant. Since the resistance is usually governed by the fixed load resistor (50 Ω), reducing the RC time constant is achieved by decreasing the device capacitance. In a p–i–n photodiode this can be achieved by minimizing the active junction area ($C_{pd} \sim A$) and maximizing the depleted absorber thickness ($C_{pd} \sim d^{-1}$), which, in turn, results in longer transit times. As a result of these considerations, the bandwidth-efficiency-product of normal-incidence p–i–n photodiodes is limited to approximately 20 GHz [7], which enables operation up to 40 Gbit/s. However, due to device miniaturization, integration of matching circuits on chip and improvements in the vertical layer structure, several very high-speed vertically-illuminated p–i–n photodiodes with acceptable quantum efficiencies have been demonstrated. Wey et al. achieved a 3 dB bandwidth of 110 GHz with a back-illuminated p–i–n PD [6]. Even though a thin absorber (200 nm) was employed the external quantum efficiency was 30 % as an

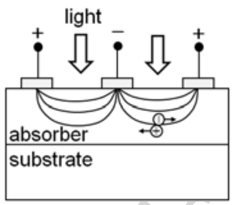


Fig. 6.3 MSM photodetector

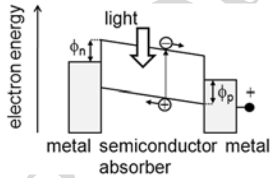


Fig. 6.4 Schematic band diagram of MSM PD

enhanced efficiency was obtained by "double pass" of the light from the top electrode (Fig. 6.2(b)). To increase the resistance-capacitance (RC)-bandwidth limitation a matched resistor (50 Ω) was integrated on chip. Hence, the effective load was reduced to 25 Ω, however, since the resistor forms a current divider with the load, half the output current was lost which resulted in a 6-dB reduction in output power at low frequencies. If a reduced spectral width can be accepted, the resonant cavity enhanced (RCE) photodetector represents another solution. With the integration of mirrors at the top and bottom of the structure multi-pass absorption can be achieved, leading to doubled quantum efficiency as reported in [8].

6.2.2 *Metal-Semiconductor-Metal Photodetector*

A metal-semiconductor-metal (MSM) photodetector consists of an undoped semiconductor absorption layer on which two interdigitated metal electrodes have been deposited. Hence, it can be basically described as two Schottky diodes connected back-to-back. For operation a voltage has to be applied to the electrodes to completely deplete the absorber and generate an electric field within the absorber (Fig. 6.3). Since one of the diodes is in reverse bias while the other is forward biased, the MSM detector exhibits symmetric current-voltage and capacitance-voltage characteristics. Figure 6.4 shows the simplified band structure of an MSM detector under bias condition. Due to the Schottky barriers, Φ_n and Φ_p, formed at the metal-semiconductor interfaces, carriers are prevented from entering the semiconductor from the metal contacts which lowers the dark current. This is in contrast to photoconductors which consist of Ohmic metal-semiconductor contacts. Under illumination photogenerated carriers drift to the electrodes and contribute to the photocurrent. Compared to p–i–n PDs this type of planar photodetector generally exhibits lower capacitance, a simple fabrication process, and is well suited for monolithic integration with metal semiconductor field transistors (MESFET) [9]. However, the responsivity of top-illuminated MSM photodetectors is significantly lower compared to p–i–n PDs due to shadowing of the electrode fingers. This issue has been addressed by using either semitransparent electrodes made, for example, from cadmium tin oxide [10] or thin metal [11], or by back-illumination [12, 13]. Using back-illumination through the substrate, the responsivity can be at least doubled [13]. Similar to p–i–n photodiodes the bandwidth of an MSM photodetector is generally governed by RC- and carrier transit time effects. Since the MSM PD has lower capacitance per unit area compared to PDs based on p–n junctions, its bandwidth is usually determined by carrier transit times. Although the optical radiation propagates perpendicular to the direction of the charge carrier transport (Fig. 6.3), the MSM PD suffers from a trade-off between the quantum efficiency and bandwidth: In order to increase the quantum efficiency of a front-illuminated MSM detector the absorber has to be made thicker and the electrode finger spacing needs to be enlarged, which, in turn, leads to an increase in carrier transit times and hence lower bandwidth. Back-illuminated MSM photodetectors are even more adversely affected by longer transit times since the carriers are predominantly generated further away from the high electric field regions near the top electrodes. Compared to their front-illuminated counterparts they achieve lower bandwidths. In some cases the bandwidth reduction is as great as 50 % [13].

In the last three decades, high-performance GaAs-based MSM photodetectors operating in the 0.85 μm-wavelength window have been extensively studied. Owing to their large Schottky barrier of 0.7 eV, these MSM photodetectors have achieved low dark currents and high speed [14]. For operation at the longer telecommunication wavelengths MSM photodetectors with a narrow-bandgap InGaAs absorber are required. However, as the Schottky barrier height of undoped InGaAs is only 0.2 eV, the direct deposition of the electrode on the InGaAs results in unacceptably large leakage currents at low bias voltage. To solve this problem a thin undoped barrier-enhancement layer (GaAs, InAlAs or InP) has been introduced between the electrode and the absorption layer [9]. Since the resulting bandgap discontinuity may be responsible for some performance degradation due to charge pile-up at the interface, additional compositionally graded layers or a graded superlattice region have been placed between the hetero-interfaces to improve device performance [15].

Due to the low capacitance of the MSM structure very large area detectors with notable bandwidth have been reported. In [16] a $350 \times 350\ \mu m^2$ MSM detector with 0.4 A/W responsivity and 900 MHz bandwidth has been achieved. A $1 \times 1\text{-mm}^2$ area MSM photodetector with 1.02 A/W responsivity at 1.53 μm wavelength and 210 MHz bandwidth was reported in [17]. To reduce carrier transit times and thus achieve higher speed the electrode finger widths and gaps were further downscaled into the sub-micron range. By applying direct electron beam lithography, front-illuminated InGaAs MSM photodetectors with 0.2 μm feature size finger electrodes with 70 GHz bandwidth have been demonstrated [18]. Recently, a compact MSM waveguide-integrated Germanium-on-insulator photodetector with 10 fF capacitance and operating speed sufficient to support data rates at 40 Gbit/s was demonstrated in [19]. The barrier height and the dark current of a 30-μm long photodetector at 1 V bias voltage were determined to be 0.08 eV and 90 μA, respectively.

6.2.3 Avalanche Photodiodes

Unlike the previously described photodiode structures, the avalanche photodiode (APD) can achieve substantially better sensitivity due to an internal gain mechanism. In an APD the photogenerated carriers are accelerated in a high-field drift region to such an extent that they generate new electron-hole pairs by impact ionization. Thus, a single photon is able to produce multiple electron-hole pairs. The multiplication factor M quantifies the photocurrent enhancement and is typically between 10 and 100 for fiber optic receivers. Although APDs require more complex epitaxial wafer structures and bias circuits, they have been successfully deployed in optical receivers that operate up to 10 Gbit/s. Due to their gain APDs provide higher sensitivity in optical receivers (PD + amplifier) than p–i–n photodiodes [20–23]. This advantage, however, applies only to thermal-noise-limited receivers since the gain mechanism is accompanied by an excess noise which depends on the carrier multiplication statistics. The excess noise also affects the speed of an APD through the avalanche build-up time which gives rise to the gain-bandwidth product of an APD.

The multiplication region of an APD plays a critical role in determining its performance, specifically the gain, the multiplication noise, and the gain-bandwidth product. According to McIntyre's local-field avalanche theory [24–26], both the noise and the gain-bandwidth product of APDs are determined by the electron-, α, and hole-, β, ionization coefficients of the semiconductor in the multiplication region, or more specifically, the ionization coefficient ratio, $k = \beta/\alpha$ if $\beta < \alpha$ and $k = \alpha/\beta$ if $\beta > \alpha$. The shot noise current for mean gain M is given by

$$\langle i^2_{shot,APD} \rangle = 2qM^2(I_{pd} + I_d)F(M)\Delta f \qquad (6.16)$$

where $F(M)$ is the excess noise factor, which arises from the random nature of impact ionization. Under the conditions of uniform electric fields and injection of the carrier with the highest ionization coefficient, the excess noise factor is

$$F(M) = kM + (1-k)\left(2 - \frac{1}{M}\right) \qquad (6.17)$$

Equation (6.17) has been derived under the condition that the ionization coefficients are in local equilibrium with the electric field, hence, the designation "local field" model. This model assumes that the ionization coefficients at a specific position are determined solely by the electric field at that position. It is clear from (6.17) that lower noise is achieved when $k \ll 1$. The gain-bandwidth product results from the time required for the avalanche process to build up or decay; the higher the gain, the higher the associated time constant and, thus, the lower the bandwidth. Emmons [27] has shown that the frequency-dependent gain can be approximated by the expression

$$M(2\pi f) = \frac{M_0}{\sqrt{1 + (2\pi f M_0 k\tau)^2}} \qquad (6.18)$$

where M_0 is the DC gain and τ is approximately (within a factor of $\sim$2) the carrier transit time across the multiplication region. It follows from this expression that for $M_0 > \alpha/\beta$ the frequency response is characterized by a constant gain bandwidth-product that increases as k decreases.

There are three documented methods to achieve low excess noise in an avalanche photodiode. The best-known approach is to select a material with low-noise characteristics, i.e., $k \ll 1$, such as Si [28–31]. Si APDs were widely used in first-generation optical fiber communication systems which operated in the wavelength range 800 nm to 900 nm [32]. Since the attenuation and dispersion characteristics of optical fibers favor operation in the now standard telecommunication wavelength range of 1.3 μm to 1.6 μm, subsequent generations employed materials appropriate for longer wavelength operation, primarily those lattice-matched to InP. One such material, $In_{0.53}Ga_{0.47}As$ (referred to in the following as InGaAs), which has a bandgap energy of 0.75 eV, is widely used as the absorbing layer in a wide range of telecommunication detectors. However, the narrow bandgap, which enables high responsivity in the short wavelength infrared (SWIR) spectrum also results in excessive dark current due to tunneling at the high electric fields required for impact ionization in InGaAs homojunctions [33, 34]. This led to the development of separate absorption and multiplication (SAM) APD structures [35]. In these APDs the p–n junction and thus the high-field multiplication region is located in a wide bandgap semiconductor such as InP where tunneling is insignificant and absorption occurs in an adjacent InGaAs layer. By properly controlling the charge density in the multiplication layer, it is possible to maintain a high enough electric field to achieve good avalanche gain while keeping the field low enough to minimize tunneling and impact ionization in the InGaAs absorber. However, the frequency response of SAM APDs, as originally implemented, was very poor owing to accumulation of photogenerated holes at the absorption/multiplication heterojunction interface [36]. To eliminate the slow release of trapped holes a transition region consisting of one or more latticed-matched, intermediate-bandgap InGaAsP layers was introduced [37, 38]. A second modification to the original SAM APD structure has been the inclusion of a high-low doping profile in the multiplication region [39–41]. In this structure the wide-bandgap multiplication region consists of a lightly doped layer where the field is high, and an adjacent, doped charge layer or field control region. This type of APD, which is frequently referred to as the SACM structure with the "C" representing the charge layer, decouples the thickness of the multiplication region from the charge density constraint in the SAM APD. Most of the initial work on InP/InGaAsP/InGaAs SAM and SACM APDs utilized mesa structures owing to their fabrication simplicity and reproducibility. However, the consensus that planar structures can effectively suppress edge breakdown spurred their development. Figure 6.5 shows a schematic cross section of an InP/InGaAsP/InGaAs SACM APD with a double diffused floating guard ring [42]. The adjacent graph shows the electric field profile normal to the surface and illustrates how the charge layer is used to tailor the relative fields in the multiplication and absorption layers.

Low excess noise and high gain-bandwidth product have also been achieved by submicron scaling of the thickness of the multiplication region, w_m. This is somewhat counterintuitive since it appears to contradict the local field model. As w_m

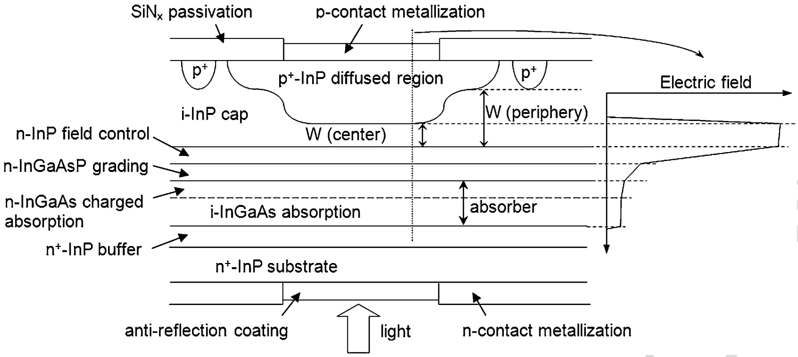
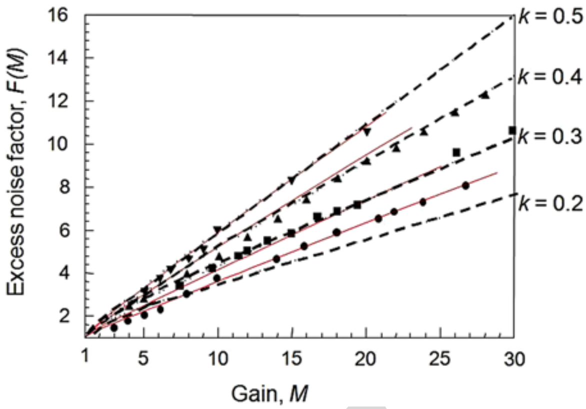


Fig. 6.5 Schematic cross section of InP/InGaAsP/InGaAs SACM APD with double-diffused floating guard ring configuration

is reduced, in order to maintain the same gain, the electric field intensity must increase in order to reduce the distance between ionization events. However, for high electric fields the electron and hole ionization coefficients tend to merge so that k approaches unity. Consequently, based on the excess noise expression in (6.17), higher excess noise would be expected for the same gain. However, in contrast to the basic assumption of the local-field model, it is well known that impact ionization is non-local in that carriers injected into the high field region are "cool" and require a certain distance to attain sufficient energy to ionize [43]. This also applies to carriers immediately after ionization because their final states are typically near the band edge. The distance in which essentially no impact ionization occurs is frequently referred to as the "dead space". If the multiplication region is thick, the dead space can be neglected and the local field model provides an accurate description of APD characteristics. However, for thin multiplication layers the non-local nature of impact ionization has a profound impact as the ionization process becomes more deterministic. Several models [44–50] have successfully been developed to accurately include the effect of the dead space. Practical noise reduction in thin APDs has been demonstrated for a wide range of materials [51–53]. Figure 6.6 shows the excess noise figure versus gain for GaAs APDs with w_m in the range from 0.1 µm to 0.8 µm [46]. The dashed lines are plots of (6.17) for $k = 0.2$ to 0.5. These lines are not representative of the actual k values; they are presented solely for reference because the k value has become a widely used indirect figure of merit for excess noise. For constant gain, it is clear that the excess noise falls significantly with decreasing w_m.

While shrinking the multiplication region thickness is an effective approach to noise reduction, it should be noted that this is relative to the characteristic noise of the bulk (thick) material. Thus, it appears that lower noise can be achieved by beginning with "low-noise" semiconductors. For this reason, InAlAs, which can be grown lattice-matched on InP substrates, is an attractive candidate for telecommunications APDs. Watanabe et al. [54] measured the ionization coefficients for InAlAs and found that $k = \beta/\alpha \sim 0.3$ to 0.4 for electric fields in the range from

Fig. 6.6 Comparison of calculated noise curves (*solid lines*) with experimental data for GaAs homojunction APD of different thickness 0.1 μm (●), 0.2 μm (■), 0.5 μm (▲), and 0.8 μm (▼) [56]



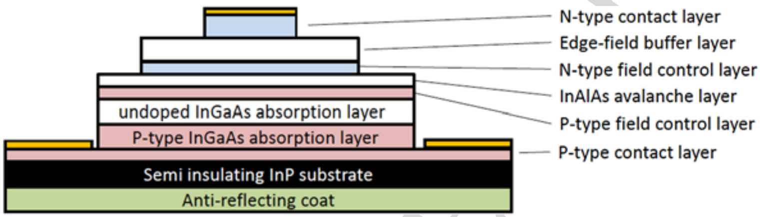


Fig. 6.7 Schematic cross-sectional view of an inverted InAlAs/InGaAs p-down APD with triple-mesa structure

400 to 650 kV/cm which compares favorably with $k = \alpha/\beta \sim 0.4$ to 0.5 for InP. Thin layers of InAlAs have also been incorporated into the multiplication region of SACM APDs. Ning Li et al. [55] reported that mesa-structure undepleted-absorber InAlAs APDs with 180 nm-thick multiplication regions exhibited excess noise equivalent to $k = 0.15$ and a gain-bandwidth product of 160 GHz. Several planar InAlAs/InGaAs SACM APDs have also been developed. An AlInAs/InGaAs planar SACM APD without a guard ring has achieved gain >40, high external quantum efficiency (88 %), 10 GHz low-gain bandwidth, and a gain-bandwidth product of 120 GHz [56, 57]. Higher operating bandwidth and gain-bandwidth product have been achieved with a triple mesa p-down InAlAs/InGaAs APD. A schematic cross section of the structure is shown in Fig. 6.7. In order to optimize the transit time, the InGaAs absorber is partially depleted similar to that in partially depleted absorber UTC-type photodiodes [58]. The 100 nm InAlAs multiplication layer enabled a gain-bandwidth of 235 GHz [59]. A triple-mesa configuration is employed to eliminate edge breakdown by reducing the electric field at the periphery of the device [60]. Devices designed for 25 Gbit/s have achieved 72 % external quantum efficiency, bandwidths of 18.5 GHz and 23 GHz at gain of $M = 10$ and 4.47, respectively, and an excess noise factor corresponding to $k = 0.2$. A four channel receiver utilizing these APDs demonstrated 100GbE over 50 km; all channels achieved receiver sensitivities better than −20 dBm at 10^{-12} BER [59].

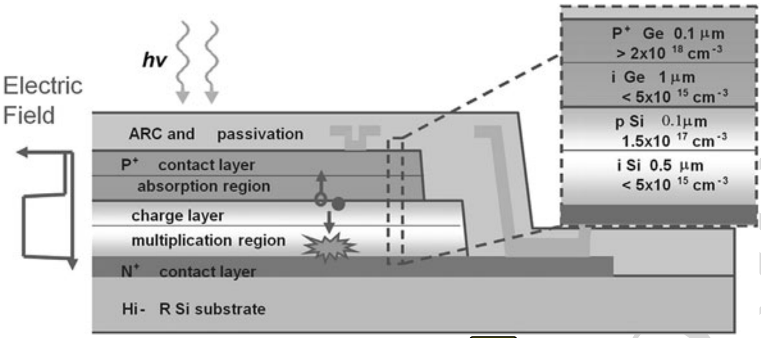


Fig. 6.8 Schematic cross section of a Ge/Si SACM APD [

The low noise of Si APDs has motivated several approaches to merge Si multiplication regions with long-wavelength absorbers. Si/InGaAs APDs have been fabricated by wafer bonding [61–63]. These APDs have achieved low dark current (4×10^{-5} A/cm^2 @ $M = 50$) [62], excess noise levels comparable to Si homojunction devices ($k \sim 0.02$) and bandwidths up to 4.8 GHz [61]. However, these APDs have not displaced InP/InGaAs SACM APDs owing to materials issues related to the bonded interface between InGaAs and Si. The SACM structure has been successfully applied to a monolithically grown Ge/Si APD in which light absorption and carrier multiplication occur inside Ge and Si, respectively [64]. A schematic cross-section of the device structure is shown in Fig. 6.8. A 1 μm-thick Ge absorber was grown on the Si multiplication layer (0.5 μm) by chemical vapor deposition. A Si charge layer was used to maintain a low electric field at the SiGe interface. Of particular note, these APDs achieved gain-bandwidth product of 340 GHz, which is two to three times higher than InP/InGaAs APDs. The effective noise factor, k, was only 0.09. Optical receivers built with these APDs demonstrated a sensitivity of -28 dBm at 10 Gbit/s and a bit error rate (BER) of 10^{-12} [64]. Using a similar structure, M. Huang et al. reported 18 GHz at a gain of 8, responsivity =0.55 A/W at 1310 nm, and successful operation at 25 Gbit/s; the receiver sensitivity was -20 dBm [65].

It has been shown that the noise of APDs with thin multiplication regions can be reduced even further by incorporating new materials and impact ionization engineering (I^{2}E) with appropriately designed heterostructures [66–73]. Structurally, I^{2}E is similar to a truncated multiple quantum well however, operationally there is a fundamental difference in that these APDs do not invoke heterojunction band discontinuities. Their function relies instead on the differences in threshold energies for impact ionization between adjacent wide-bandgap and narrower-bandgap materials. The structures that have achieved the lowest excess noise, to date, utilize multiplication regions in which electrons are injected from a wide bandgap semiconductor into adjacent low bandgap material. Recently, InGaAlAs/InP implementations that operate at the telecommunications wavelengths have been reported. Using both a single-well structure and a pseudo-graded bandgap based on InAlAs/InGaAlAs materials Wang et al. [72] demonstrated excess noise equivalent to $k \sim 0.12$ and dark

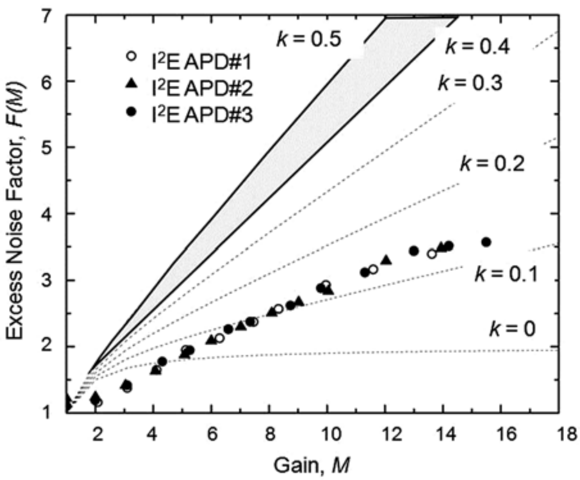


Fig. 6.9 Excess noise factor, $F(M)$, versus gain for an SACM APD with I^{2}E $In_{0.52}Ga_{0.15}Al_{0.33}As/In_{0.52}Al_{0.48}As$ multiplication region [ ]

current comparable to that of homojunction InAlAs APDs. Duan et al. have incorporated a similar I^{2}E multiplication region into an MBE-grown InGaAlAs I^{2}E SACM APD [73]. Based on Monte Carlo simulations of similar GaAs/AlGaAs I^{2}E APDs [69], it can be inferred that there are relatively few ionization events in the $In_{0.52}Al_{0.48}As$ layer, owing to the combined effects of "dead space" and the higher threshold energy in $In_{0.52}Al_{0.48}As$. Figure 6.9 shows the excess noise factor, $F(M)$, versus gain. The dotted lines in Fig. 6.8 are plots of $F(M)$ for $k = 0$ to 0.5. For $M \leq 4$, it appears that $k < 0$, which is unphysical and simply reflects the inapplicability of the local field model for this type of multiplication region. At higher gain, the excess noise is equivalent to a k value of ~0.12. For reference, the excess noise factor for $InP/In_{0.53}Ga_{0.47}As$ SACM APDs is shown as the shaded region in Fig. 6.9.

Interest in extending the transmission of optical communications into the 2 µm band has been stimulated by the development of hollow core photonic bandgap fibers. Detectors that operate in the mid-wave infrared (MWIR) spectrum are not as advanced as those in the SWIR. A promising candidate for 2 µm APDs utilizes InAs as the gain material. InAs APDs have also demonstrated $k \sim 0$ with moderately low dark current at room temperature [74, 75]. InAs APDs employing 10 µm-thick intrinsic regions and AlAsSb blocking layer to suppress electron diffusion current achieved gain as high as ~300 at 15 V bias. The gain-bandwidth product was >500 GHz, however, the bandwidth was transit-time limited in the range 2 to 3 GHz independent of gain owing to the thick depletion width required to minimize the tunneling component of the dark current.

6.2.4 Advanced Photodiode Structures

Driven by the requirements of coherent fiber optic links and high-speed analog systems there has been increased interest in high-speed photodiodes that achieve the requisite bandwidths without sacrificing responsivity, linearity, or output power signal levels [76–78]. In analog fiber optic links high output photocurrent levels help to minimize the noise figure and hence increase the dynamic range [79, 80]. However, there are several physical mechanisms that impact saturation in photodiodes, including space-charge screening [81, 82], and thermal [83, 84] effects. The space charge effect has its origin in the spatial distribution of the photogenerated carriers as they transit the depletion layer. At high current densities, as electrons and holes travel in opposite directions, an internal space-charge field is generated that opposes the bias electric field. For sufficiently high optical input power levels, the space-charge induced electric field can be strong enough to make the bias electric field collapse, which will result in reduced carrier drift velocities, longer transit times, and hence RF photocurrent compression [85]. Higher applied voltages help to mitigate these effects, however, eventually the added Joule heating increases junction temperatures which can cause device failure. The thermal limit is determined by the heat dissipation characteristics of the constituent semiconductor layers, the photodiode geometry, and by the heat sink design. Joule heating can result in temperatures as high as 500 °C in the depletion region [84] which can cause device thermal and/or electrical failure. When large output photocurrent is delivered to the load, the voltage drop across the load and the device series resistance can effectively remove the available voltage bias from the depletion region and thus negatively impact the photodiode saturation. A measure of the photodiode saturation is the saturation current which is defined as the average photocurrent at which the electrical output power at the cut-off frequency deviates by -1 dB from an ideal current-power relation at the load resistor.

To address the space-charge effect several photodiode structures have been developed including the dual-depletion region (DDR) photodiode [86], the uni-traveling carrier (UTC) photodiode [87, 88] and the partially-depleted-absorber (PDA) photodiode [89].

The DDR PD is characterized by a transparent drift layer between the intrinsic absorber and the n-type contact layer (Fig. 6.10(a)). Compared to a p–i–n PD with the same depleted absorber thickness it allows for a reduced junction capacitance for the same hole transit time. In the structure the photogenerated holes transit only the InGaAs absorbing layer whereas the electrons travel across both the absorbing layer and the InP drift layer. Since electrons are faster than holes, the overall bandwidth in otherwise RC-limited devices can be reasonably increased when the layer thicknesses are designed properly. To date, DDR PDs with bandwidths up to 50 GHz and 0.7 A/W responsivity have been published [90]. Using a graded-index (GRIN) lens for uniform illumination, which mitigates the space-charge effect [81], saturation photocurrents of 45 mA at 10 GHz were achieved [91].

The band diagram of the UTC PD is schematically shown in Fig. 6.10(b). The active part of the UTC PD consists of a p-type narrow-bandgap light absorption layer and an undoped, wide-bandgap (transparent) depleted carrier-collection layer. The photogenerated minority electrons in the neutral absorption layer are transported by diffusion and/or drift into the depleted collection layer. In order to accelerate the electron diffusion process, a built-in electric field can be generated in the p-doped absorber by well-controlled bandgap- or doping-grading. Once the electrons

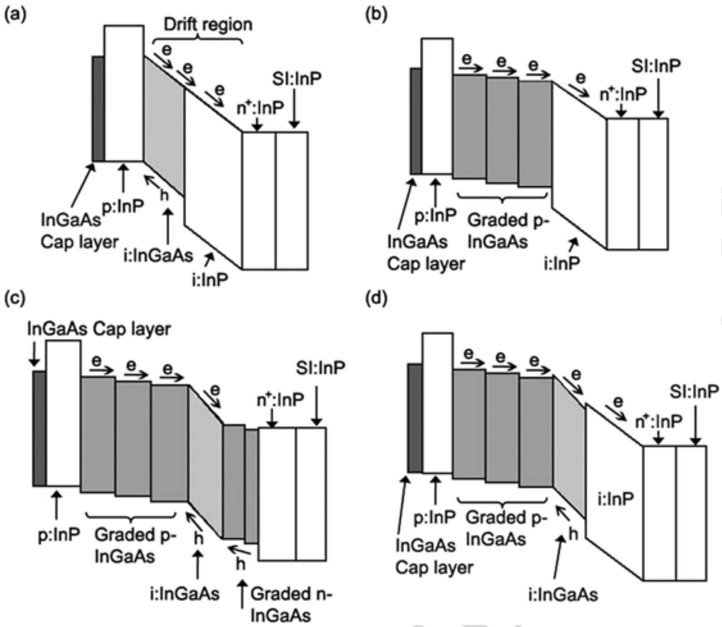


Fig. 6.10 Schematic band diagrams of (**a**) DDR PD, (**b**) UTC PD, (**c**) PDA PD and (**d**) MUTC PD with InGaAs absorbing layers and InP contact layers on semi-insulating (SI) InP substrate. The transit paths of photogenerated electrons (e) and holes (h) are indicated

reach the high-field collection layer, they drift toward the n-contact at a high saturation velocity. On the other hand, since the absorption layer is quasi-neutral, photogenerated majority holes respond very fast within the dielectric relaxation time by their collective motion. This is an essential difference from the conventional p–i–n PD, in which both electrons and holes contribute to the response current and the low-velocity hole-transport dominates the speed performance and exacerbates the space-charge effect [92]. Since electrons maintain their high velocity at relatively low electric fields, the UTC PD can achieve high speed and high saturation output photocurrent even at low bias voltage [93].

In the UTC PD the transit time for electrons has two components corresponding to the undepleted absorber and the drift region with widths W_A, and W_C, respectively. The total transit-time is [94]:

$$\tau_i = \frac{W_C}{3.5\bar{v}} + \left[\frac{W_A^2}{3D_e} + \frac{W_A}{v_{th}}\right]$$

where D_e is the electron diffusion coefficient and v_{th} is the electron thermal velocity. $W_A^2/3D_e$ is the diffusion transit time and W_A/v_{th} is the correction factor associated with the finite thermal velocity.

The reduction of absorber and drift region widths and the miniaturization of the active area has led to UTC photodiodes that achieved high saturation current and very high-speed. Shimizu et al. reported a 20-μm^2 back-illuminated UTC PD with a 3 dB bandwidth of 152 GHz, an output peak voltage of 0.7 V, and 13 % external quantum efficiency [95]. A UTC PD with bandwidth of 220 GHz at 25 Ω effective load and 0.13 A/W was presented in [96]. In addition, the carrier transit time in UTC photodiodes can be further reduced by exploiting the velocity overshoot of electrons in the depletion layer [97]. By introducing an additional p-type charge layer between collector and n-contact layer, the electric field inside the structure can be adjusted to benefit from the electron velocity overshoot [98]. Using this structure Wu et al. have reported a 64 μm^2 device having a 3 dB bandwidth of 120 GHz at 25 Ω effective load. The back-illuminated PD with an integrated micro-lens on the substrate showed a responsivity of 0.15 A/W [98]. Recently, a similar device flip-chip bonded onto an AlN substrate for improved heat sinking achieved a saturation photocurrent of 37 mA [99].

In [58] Li et al. proposed a structure where a thin InGaAs depletion layer was combined with undepleted InGaAs absorbers to increase responsivity (Fig. 6.10(c)). This structure is called a partially-depleted-absorber (PDA) photodiode and provides higher responsivity than a UTC PD with the same undepleted absorber thickness. Balancing the hole and electron densities within the depletion region was previously suggested as a technique for minimizing the space charge effect and for increasing photocurrents [85]. In the PDA, photodiode charge balance is accomplished by the p-doped absorber and an n-doped absorber on each side of the i-region. The p-doped absorber injects electrons into the i-region while the n-doped absorber injects holes. In the implementation, electron injection is stronger than that of holes due to the different thicknesses of the absorbers on each side of the i-layer. Since thin depletion layers are essential to obtain high currents, the PDA PD was designed with a thinned i-layer (250 nm) which reduces space-charge screening and minimizes thermal effects across the depletion layer [92]. Due to the poor thermal conductivity of InGaAs high-power InP/InGaAs PDs need to avoid thick InGaAs depletion layers to prevent thermally-induced degradation and potential device failure at high optical photocurrents [85]. The measured compression photocurrent of a back-illuminated 8 μm-diameter PDA PD was 24 mA at 48 GHz with responsivity of 0.6 A/W [58].

A hybrid structure of UTC and PDA has been proposed by Jun et al. in [100]. This structure, called a modified UTC (MUTC), is formed by inserting an undoped i-InGaAs layer between the p-type InGaAs absorber and the InP drift layer (Fig. 6.10(d)). Thus the MUTC PD provides an additional design parameter which allows higher responsivity and higher bandwidth when the layer design is optimized [101]. The epitaxial layer structure of an MUTC PD that reached very high saturation current is shown in Fig. 6.11 [102]. The InGaAs absorber region was comprised of a 150 nm-thick depleted layer and 700 nm step-graded p-doped layers. The latter create a quasielectric field that enhances the electron diffusion toward the drift region [95]. The 900 nm InP electron drift layer was slightly n-type doped for space charge compensation [103]. The incorporated positive charges pre-distort the electric field to partially compensate the field change caused by the

Layer	Material	Doping (cm^{-3})	Thickness (nm)
P-contact	$InGa_{0.47}As_{0.53}$	p^+ doping, 2×10^{19}	50
Block layer	InP	p^+ doping, 1.5×10^{18}	100
Quaternary layer	InGaAsP, Q 1.1	p^+ doping, 2×10^{18}	15
Quaternary layer	InGaAsP, Q 1.4	p^+ doping, 2×10^{18}	15
Graded doped absorber	$InGa_{0.47}As_{0.53}$	p^+ doping, $5 \times 10^{17} \sim 2 \times 10^{18}$	700
Depleted absorber	$InGa_{0.47}As_{0.53}$	n^- doping, 1×10^{16}	150
Quaternary layer	InGaAsP, Q 1.4	n^- doping, 1×10^{16}	15
Quaternary layer	InGaAsP, Q 1.1	n^- doping, 1×10^{16}	15
Cliff layer	InP	n doping, 1.4×10^{17}	50
Drift layer	InP	n^- doping, 1×10^{16}	900
N-contact	InP	n^+ doping, 1×10^{19}	1000
Substrate	InP, semi-insulating, double side polished		

Fig. 6.11 Epitaxial layer structure of charge-compensated MUTC photodiode [106]

space charge in the presence of high photocurrents. A "cliff" layer was incorporated between the absorber and charge-compensated transparent drift layers to enhance the electric field in the depleted absorber layer [104, 105]. To improve thermal dissipation these MUTC-PDs were flip-chip bonded onto high-thermal conductivity submounts. Using an Au–Au thermo-compression bonding process the maximum dissipated DC power ($I_{pd} \times V_{bias}$) of the photodiodes was increased by up to 90 % as the result of flip-chip bonding onto AlN substrate [106]. Even higher values were obtained when using diamond substrates. Photodiodes with diameters of 28 µm and 50 µm achieved RF output powers of 26 dBm at 25 GHz, and 32.7 dBm at 10 GHz, respectively, the average photocurrent reached 300 mA [102]. Recently, device miniaturization and optimization of the on-chip microwave transmission lines enabled MUTC PDs with 3-dB bandwidths up to 65 GHz and 16 dBm RF output power [107]. In this design an air-bridge connected the photodiode to a high-impedance transmission line (∼85 Ω) which was also the bond pad in the flip-chip-bonding process. As the transmission line was designed to provide slight inductive peaking, the bandwidth was expanded by 30 % beyond the conventional RC-limitation.

6.2.5 High-Speed Side-Illuminated Photodiodes

To overcome the bandwidth-efficiency trade-off side-illuminated waveguide-photodiodes (WGPD) have been developed for p–i–n [108, 109], MSM [18, 19], UTC [110–113], PDA [114], and APD photodiodes [115–118]. WGPD structures are illustrated in Fig. 6.12. The primary benefit of this type of photodetector is that

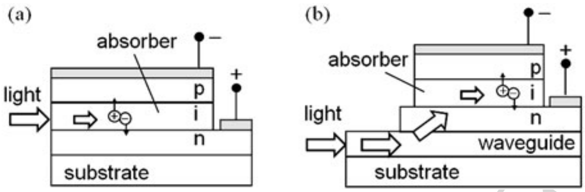


Fig. 6.12 Side-illuminated (**a**) WGPD and (**b**) evanescently-coupled WGPD

high efficiency and short carrier transit times can be achieved simultaneously [119]. Waveguide-photodiodes utilize an input optical waveguide with embedded absorbing layer. The photogenerated carriers transit only the thin absorption/depletion region perpendicular to the epitaxial layers which enables high bandwidths. Since electrical and optical transports are not collinear, the carrier transit times are determined by the thickness of the absorber while the length of the detector primarily controls the responsivity. Assuming a single mode WGPD, the responsivity is given by [7]:

$$R_{WGPD} = R_{ideal}(1 - R_0)\eta_c\left[1 - \exp(-\Gamma_{xy}\alpha l_{abs})\right] \tag{6.19}$$

Here, η_c is the input coupling efficiency determined from the overlap integral between the optical fields of the input (fiber) and WGPD, Γ_{xy} is the confinement factor in the xy-plane (perpendicular to the propagation direction) which quantifies the fraction of power confined within the absorbing layer, and l_{abs} is the PD length. For high-speed WGPDs with thin absorbers (<1 μm) either η_c or Γ_{xy} is <1. This is because the diameter of the fiber input light spot, even when focused by lenses, is no less than 2 μm, while the optical field distribution well confined within the absorbing layer is narrower than 1 μm. On the other hand, having a wider optical field distribution in the WG-PD improves η_c, however, at the expense of a reduced Γ_{xy}.

Using a double core, i.e. large multimode optical cavity, Kato et al. presented a WGPD with a record bandwidth-efficiency product of 55 GHz in 1994 [120]. The 12 μm-long p–i–n WGPD achieved a quantum efficiency as high as 50 %. The mushroom mesa approach developed for this photodiode yielded low resistance (10 Ω) while minimizing the p–n junction area to realize very low capacitance (∼15 fF) which enabled an RC-limited bandwidth of 110 GHz.

The disadvantages of WGPDs are the limited high-power capability due to non-uniform carrier distribution along the optical path and the low tolerance to lateral and vertical displacement of the input signal. The latter implies the use of additional optics or a tapered fiber in order to efficiently illuminate the small active region. Thus, in general, a low tolerance to lateral and vertical displacement of the input signal is obtained which makes the fiber-chip coupling more crucial and complex. Recently, coupling tolerances at 1 dB extra loss of ±0.85 μm and ±1.3 μm in the vertical and horizontal direction, respectively, were reported for a 5 μm-wide tapered waveguide photodiode [121]. This photodiode had an absorption layer of only 100 nm thickness embedded within thicker depleted transparent layers to reduce capacitance and balance electron and hole drift times. The reported bandwidth was 42 GHz with responsivity of 1.08 A/W. In another approach Fukano et al. proposed the edge-illuminated refracting facet photodiode (RFPD) in which the incident light that is parallel to the top surface is refracted at an angled facet and transits the absorption layer with a certain refracted angle [122]. With this design, the fiber-chip −1 dB-misalignment tolerances in the horizontal and vertical direction were as large as 13.4 μm and 3.3 μm, respectively. Furthermore, since this design leads to an increased absorption length compared to a vertically-illuminated photodiode, a high responsivity of 1 A/W with <0.3 dB polarization dependent loss (PDL) was measured for a p–i–n RFPD with a 1-μm thick absorber [123]. Applying the UTC structure to the RFPD and further downscaling of the active area has led to high-speed, high-power UTC PDs with 3 dB bandwidths up to 310 GHz (at 12.5 Ω effective load) [124]. In [125] a refracting-facet UTC PD module with responsivity of 0.21 A/W and 0.5 dB PDL for the detection of 100 and 160 Gbit/s return-to-zero (RZ) data rates was reported. Similar detector modules have been further optimized for W-band (75–110 GHz) [126], F-band (90–140 GHz) [127] and D-band (110–170 GHz) [128] operation; the latter exhibited a maximum RF output power of 2 dBm at 150 GHz.

In contrast to these side-illuminated devices the evanescently-coupled WGPD consists of a photodiode located on top of a passive waveguide. In the structure shown in Fig. 6.12(b) the light couples evanescently from a single mode input waveguide to the PD mesa which ensures a more uniform absorption along the device length and leads to an improved high-power capability [129]. This WGPD is well-suited for the monolithic integration with additional components, such as planar lightwave circuits resulting in advanced detector structures with increased functionality or photonic integrated circuits [130]. Furthermore, independent of the active device, a mode field transformer (taper) can be integrated in order to improve the fiber-chip coupling efficiency. This enables the use of a cleaved fiber instead of a tapered/lensed fiber, which simplifies the fiber-chip coupling process and also provides large alignment tolerances of ±2.5 μm and ±3.5 μm in the vertical and horizontal directions, respectively [108]. A highly efficient waveguide-integrated p–i–n photodetector was reported in [131]. The photodetector chip comprised a p–i–n photodiode with an active area of 5×20 μm^2 and an InGaAsP/InGaAs heterostructure absorption layer stack, a vertically tapered mode field transformer, a biasing network, and a 50 Ω load resistor. An optimized impedance of the electrical output line of the detector led to an increase of the cut-off frequency to >100 GHz. Figure 6.13 shows (a) a schematic of this photodiode structure, (b) the p–i–n layer stack, and (c) the monolithically integrated bias circuitry. The chip was assembled into a package equipped with a 1 mm coaxial output connector and a fiber pigtail (Fig. 6.14, inset). Figure 6.14 shows the calibrated frequency response of the photodetector module at −2 V bias. A 3 dB bandwidth of 100 GHz with a maximum RF output power of −7 dBm was measured with an optical heterodyne setup [132]. The responsivity was 0.73 A/W.

Similar PD modules have been evaluated in several back-to-back transmission experiments [133]. Figure 6.15 shows the received electrical 80 Gbit/s return-to-zero (RZ) eye patterns at different optical input power levels using a 70 GHz-sampling oscilloscope. All measured eye patterns exhibit wide opening with a peak voltage up to 0.6 V revealing only negligible saturation effects at +12 dBm. Figure 6.16 shows

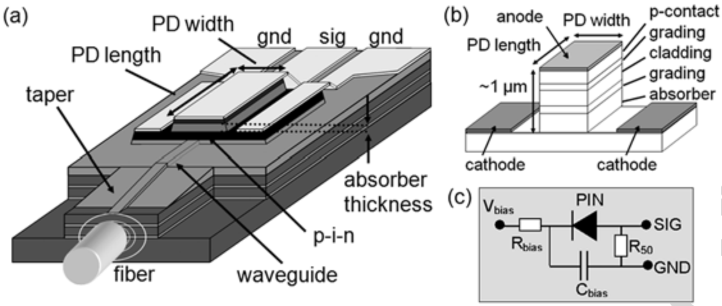


Fig. 6.13 (**a**) Evanescently-coupled WGPD, (**b**) p–i–n mesa, and (**c**) monolithically integrated bias circuitry

Fig. 6.14 Relative frequency response of the PD module (+2.3 dBm optical input power), *inset*: photograph of the PD module

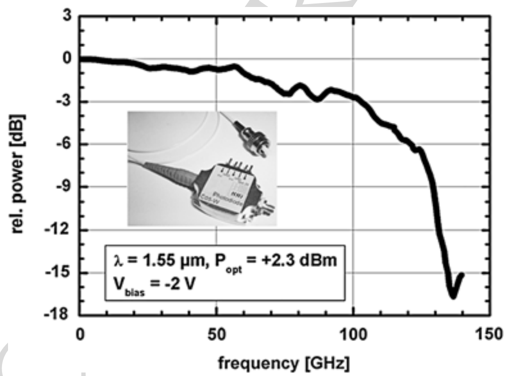
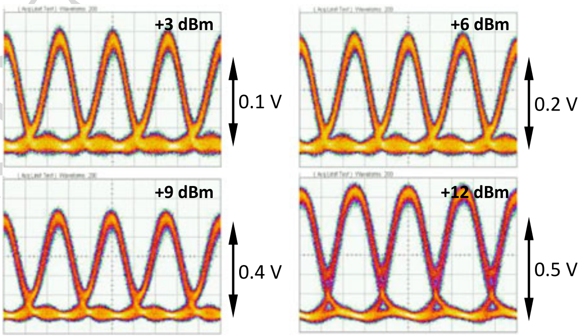


Fig. 6.15 Electrical 80 Gbit/s return-to-zero (RZ) eye pattern at 3, 6, 9 and 12 dBm optical input power detected by the PD module at −2.5 V bias (x: 5ps/div)



the detected 160 Gbit/s RZ data stream at +12 dBm optical input power. Due to the insufficient bandwidth of the sampling head and the PD module an RZ-to-NRZ conversion can be observed. Nevertheless, the eye amplitude is still notable and the inner eye opening reached 160 mV.

A fully packaged photodiode of this type capable of providing a flexible DC offset voltage at its RF output achieved a bandwidth of 90 GHz. The on-chip bias

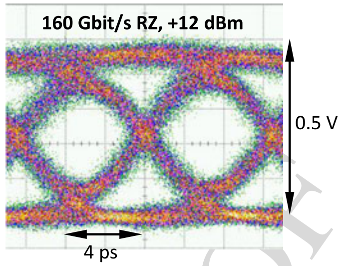


Fig. 6.16 Detected eye pattern under 160 Gbit/s RZ excitation

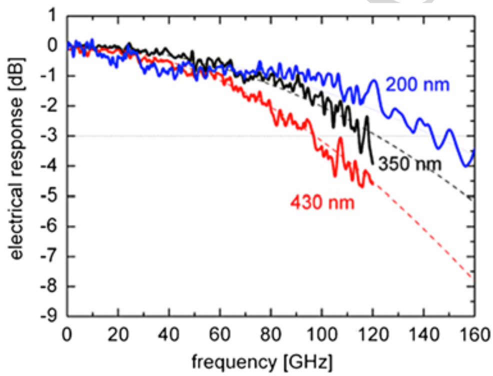


Fig. 6.17 Measured frequency responses for $5 \times 7\ \mu m^2$ p–i–n photodiodes with 430 nm, 350 nm, and 200 nm thick absorbers

network of this photodetector was modified to supply a voltage to the postamplifier or demux IC without the need of an external bias tee [134]. The responsivity of the photodetector module was 0.53 A/W with a PDL of only 0.1 dB.

The reduction of absorber thickness and PD length allowed to further enhance the bandwidth of similar p–i–n photodiodes in Refs. [135, 136]. By downscaling the PD length to 7 μm the reduced capacitance enabled transit-time limited bandwidths at 25 Ω effective load of 100 GHz, 120 GHz and 145 GHz for 430 nm-, 350 nm- and 200 nm-thick absorbers, respectively (Fig. 6.17). Owing to an optimized evanescent coupling scheme the fiber-coupled responsivities were 0.51 A/W, 0.48 A/W and 0.35 A/W, respectively. To avoid a decrease in responsivity, which generally follows the reduction in PD length, the n-contact layer was extended by a well-defined length L toward the single mode input waveguide (Fig. 6.18). Since both, the PD mesa and the protruding section L form multimode waveguides, mode beating effects can be exploited in order to facilitate an efficient coupling from the single mode waveguide into the absorber [137, 138], and thus increase Γ_{xy}. Since Γ_{xy} is a function of z (in propagation direction) and L, the responsivity of the evanescently coupled WGPD, R_{eWGPD}, can be estimated using the relation:

$$R_{eWGPD} = R_{ideal}(1 - R_0)\eta_c\left[1 - \exp\left(-\alpha \int_0^{l_{abs}} \Gamma_{xy}(z, L)dz\right)\right] \tag{6.20}$$

where the input waveguides are assumed to be lossless. It should be noted that designing a WGPD with a non-constant confinement factor can be beneficial for

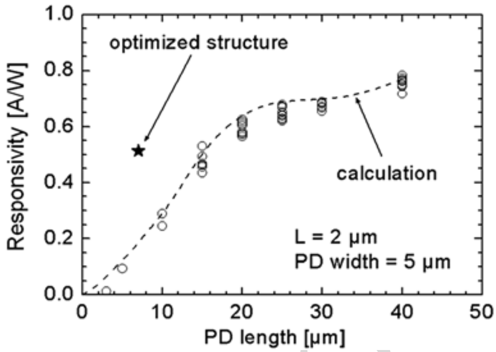


Fig. 6.18 Responsivity vs. PD length. Each *circle* displays experimental data for a different device from a previously fabricated wafer with $L = 2$ μm. The *star* indicates the optimized design with $L = 7$ μm

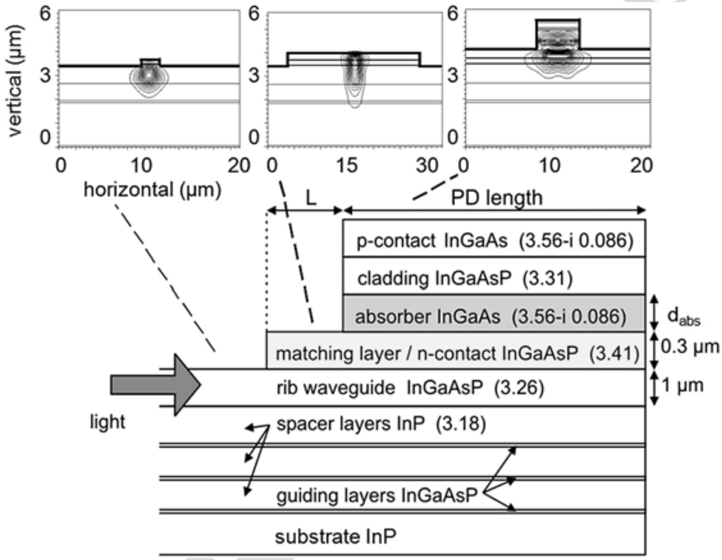


Fig. 6.19 Cross-sectional view of the PD with extended n-contact layer (matching layer). All refractive indices at 1.55 μm wavelength. The light is injected from the left into the mode field transformer (not shown) and couples evanescently from the semi-insulating waveguide into the p–i–n mesa. The *insets* depict the optical 2D field intensity profiles calculated in the single mode waveguide (*left*), multimode matching layer (*center*) and multimode PD mesa (*right*)

high-power applications [139]. To achieve a more uniform absorption profile and thus current density the confinement factor should increase toward the end of the WGPD.

Numerical simulations of the optimized structure showed that illumination, and thus absorption, are strong in the first few microns of the absorber (Fig. 6.19, insets) [140]. For maximum responsivity the protrusion length L can be estimated from the beat length of the two most prominent modes in the multimode structure reduced by the PD length. Experimentally $L = 7$ μm was found for a 7 μm-long PD which is in good agreement with the calculation using a numerically determined beat length

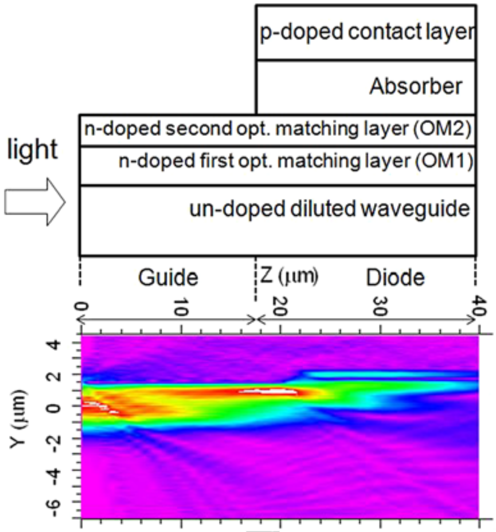


Fig. 6.20 Schematic of photodiode with planar multimode input waveguide and two optical matching layers. The *lower image* is the simulated optical intensity as light propagates through the integrated waveguide/PD structure

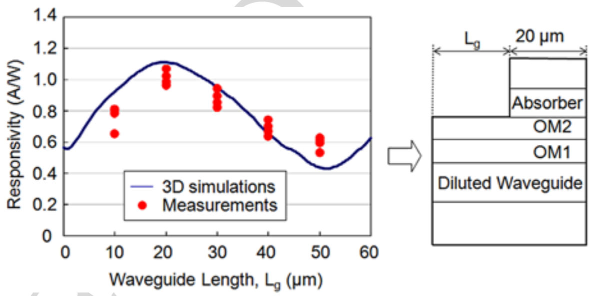


Fig. 6.21 Responsivity of photodiode with planar multimode input waveguide versus guide length for a 20-µm long active region: the *solid line* is the simulated responsivity and the *filled circles* are measurements

of 13 µm. Compared to previous devices with $L = 2$ µm a twofold responsivity increase to 0.51 A/W was reached with the optimized structure (Fig. 6.18).

It has been shown that high fiber-coupled efficiencies can be achieved without the need of a mode field transformer if a short multimode input waveguide is used [141]. The photodiode in [142] utilized a planar diluted input waveguide and two optical matching layers designed to provide a gradual increase of the optical refractive index from the diluted waveguide to the absorbing layer which resulted in a significant enhancement in the quantum efficiency. Figure 6.20 depicts a schematic of an evanescently-coupled photodiode that utilizes a planar diluted waveguide and two optical matching layers. The diluted waveguide is a stack of 10-periods of undoped InP/InGaAsP (1.1-µm bandgap) layers. The number of periods was optimized to achieve high coupling efficiency with an input fiber and low TE/TM polarization dependence. The two optical matching layers are n-doped InGaAsP with bandgaps corresponding to 1.1 µm and 1.4 µm for the first and second optical matching layers, respectively. For this approach, since the waveguide to photodiode coupling is based on mode beating effects, the coupling efficiency oscillates along the propagation direction. The solid line in Fig. 6.21 presents the responsivity simulation of 20-µm

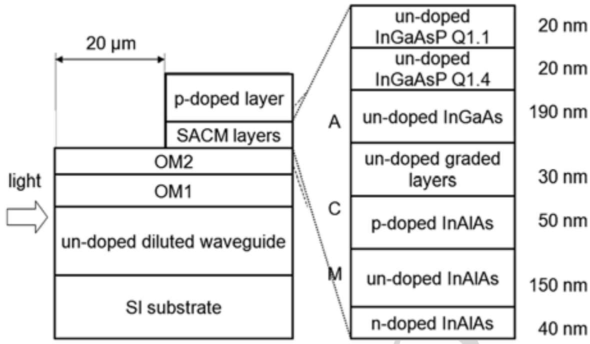


Fig. 6.22 Schematic cross section of evanescently-coupled InAlAs/InGaAs waveguide APD

long photodiodes versus the input waveguide length. Oscillations related to intermodal interferences are clearly visible in this figure. In agreement with the modeling, the responsivity was 1.07 A/W for an optimal input waveguide length when using a lensed fiber. The reported bandwidth was 48 GHz. It has been demonstrated that the length of the planar multimode waveguide can be controlled precisely by dry etching of the waveguide input facet. In addition, it was found that etching a lensed facet improved horizontal fiber-chip 1-dB alignment tolerances to >20 µm [143]. The reported lensed facet waveguide UTC PDs achieved 0.55 A/W and a bandwidth of >50 GHz [144].

Lateral tapering of diluted input waveguides was demonstrated in [145] using an asymmetric twin-waveguide technology [146]. In this PD the incident light is collected by a single mode diluted waveguide and transferred via a taper to a thinner coupling waveguide from where it couples evanescently into the absorber. Using a lensed fiber for input coupling a responsivity of ~1 A/W and a bandwidth 42 GHz were demonstrated. Using a similar waveguide taper Rouvalis et al. reported UTC waveguide PDs with a bandwidth of 110 GHz and 0.32 A/W [147]. The PD active area and absorber thickness were $4 \times 15\ \mu m^2$ and 70 nm, respectively. For sub-THz applications similar PDs with traveling wave electrodes and integrated resonant antennas achieved a maximum extracted power of 150 µW around 460 GHz.

The WGPD approach has also been applied to APDs. Demiguel et al. [148] have reported an evanescently coupled $In_{0.52}Al_{0.48}As/In_{0.53}Ga_{0.47}As$ SACM APD having a planar short multimode input waveguide. A schematic cross section of this APD is shown in Fig. 6.22. The input diluted waveguide is similar to that reported for p–i–n PDs in [142]. The photocurrent, dark current, and gain versus reverse bias are plotted in Fig. 6.23. The breakdown occurred at ~18.5 V and the dark current at 90 % of the breakdown was in the range 100 to 500 nA. The responsivity was 0.62 A/W with a PDL <0.5 dB. Figure 6.24 shows the bandwidth versus gain; at low gain the maximum bandwidth was 35 GHz and the high-gain response exhibits a gain-bandwidth product of 160 GHz. Nakata et al. [116] have reported an edge-coupled InAlAs/ InGaAs APD that achieved 0.73 A/W responsivity, low-gain bandwidth of 35 GHz, and 140 GHz gain-bandwidth product. A similar waveguide APD with a gain-bandwidth product of 170 GHz and a minimum received power of −19.6 dBm at 40 Gbit/s (for BER of 10^{-9}) was recently reported in [149].

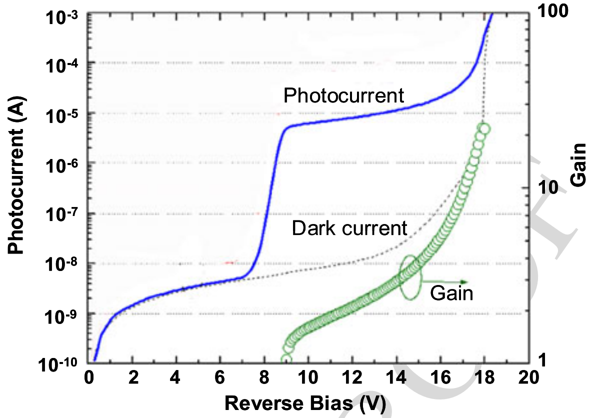


Fig. 6.23 Photocurrent, dark current, and gain versus reverse bias of evanescently-coupled InAlAs/InGaAs waveguide APD

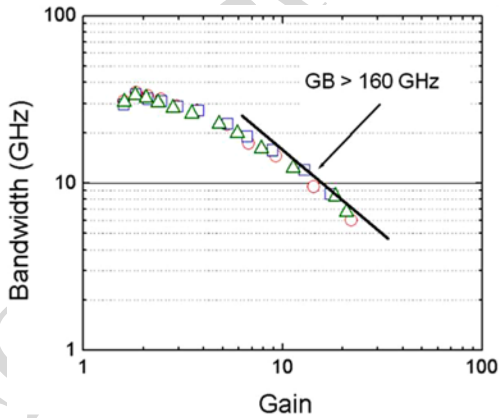


Fig. 6.24 Bandwidth versus gain of evanescently-coupled InAlAs/InGaAs waveguide APD

6.2.6 Material Systems

The group III–V semiconductor InP with its lattice-matched compounds continues to be a leading material system for the fabrication of high-performance photodiodes in the C and L bands. Owing to its high absorption efficiency, high carrier drift velocities, and good material quality (low dark current) the InGaAs/InP PD has become a standard solution for today's high-speed applications.

Varying compositions of the alloys (bandgap engineering) allow the bandgap to be varied between 0.75 eV (1.65 μm) and 1.35 eV (0.92 μm). Thus the material system allows a composition of highly absorbing and transparent layers at telecommunication wavelengths. Another advantage of the InP material system arises from the potential for monolithic integration. To date monolithic integration of all active and passive optical functions and electronic devices has been demonstrated. Hence, InP is a leading platform for photonic integrated circuits (PICs) which have the potential to enhance performance, reduce footprint, and decrease packaging costs of complex photonic devices. Due to steady progress in the field of components and manufacturing, large-scale PICs have reached sufficient reliability for deployment in optical networks [150].

However, with the rapid progress in Silicon photonics there has been increased research effort toward Si-compatible waveguide photodiodes that operate at the telecommunication wavelengths. Although Si is transparent at wavelengths >1.1 μm numerous approaches have been reported to enable efficient light detection at 1.55 μm wavelength, including ion-implanted all-silicon [151], InGaAs/GaAs growth on Si [152, 153], polycrystalline Ge films [154, 155], Si–Ge hetero-epitaxy [156–158], and III–V on Si bonding [159].

6.2.7 Germanium Photodiodes on Silicon

Since Ge can be grown on silicon substrates, this approach ultimately promises large-scale photonic-electronic integration using the available CMOS infrastructure [160–162]. To reduce defect densities arising from the $\sim$4 % lattice mismatch between Ge and Si various growth techniques including the deposition of graded SiGe buffer layers [163], high/low temperature growth [157], area-selective growth [164, 165], and cyclic annealing have been successfully developed [166]. In [161], surface-normal PDs with a bandwidth of 36 GHz and 0.47 A/W responsivity achieved a low dark current of <100 nA. Using a two-step MBE growth technique consisting of a silicon buffer followed by a very thin Ge virtual substrate grown at low temperature for lattice mismatch accommodation low-dark-current, high-speed photodiodes were demonstrated in [167]. These n–i–p photodiodes with 10 μm-diameter achieved a high bandwidth of 49 GHz and 0.05 A/W responsivity at 2 V. The absorption coefficient of Ge depends on the growth conditions, and typical values range between 1000 cm^{-1} and 4000 cm^{-1} at 1.55 μm wavelength [168, 169]. Thus, when compared to InGaAs/InP photodiodes, surface-normal Ge photodiodes typically exhibit lower bandwidth-efficiency products. Waveguide structures have the potential to achieve larger bandwidth-efficiency products and to date several high-performance Ge waveguide PDs have been demonstrated [170, 171]. Due to their nature, they have become key devices in silicon-on-insulator (SOI) photonic integrated circuits [172]. In [19], a high-speed CMOS-compatible evanescently-coupled Ge MSM waveguide photodetector was reported. In this device a 150 nm-thick Ge layer served as the absorption layer and the MSM detector was formed by implementing 150 nm-wide inter-digitated metal fingers on top of the waveguide. The dark current and the internal responsivity at 1.3 μm wavelength were 90 μA and 0.42 A/W at 1 V, respectively. The bandwidth was 40 GHz due to the detector's small active area of 0.7×20 μm^2 [19]. An even higher 3 dB bandwidth of 45 GHz was achieved with an n–i–p Ge waveguide photodiode with an active area of only 1.3×4 μm^2 in [173]. The dark current and responsivity were 3 nA at 1 V and 0.6 A/W at 1.55 μm, respectively. Recently, a lateral Ge photodetector with a large cross-section SOI waveguide was demonstrated in [174]. In this design an intrinsic Ge layer with a width of 650 nm was butt-coupled to a tapered SOI waveguide that

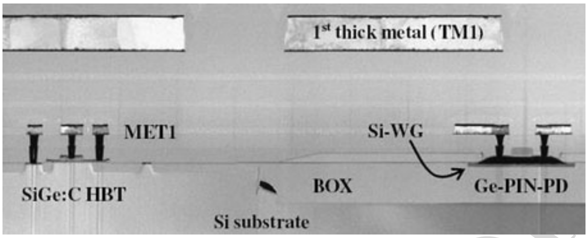


Fig. 6.25 TEM cross section of a Ge-photodiode on SOI and a SiGe:C HBT fabricated in an adjacent bulk region

enabled low-loss input coupling to the fiber. The demonstrated photodiode reached a bandwidth of 32 GHz and responsivity of 0.8 A/W at 1.55 μm wavelength. The dark current was 1.3 μA at 1 V. In [175], Vivien et al. reported a butt-coupled lateral p–i–n photodiode with >110 GHz bandwidth. At zero bias the measured responsivity was 0.8 A/W. The device was fabricated using a silicon recess etch followed by a selective Ge-regrowth. Dopants were implanted to form a horizontal p–i–n junction with a nominal intrinsic Ge width of 500 nm. Recently, Liow et al. demonstrated a waveguide photodiode with a high responsivity of 1.3 A/W at 1.61 μm wavelength [176]. In their experiments they applied a bias voltage of 9 V to achieve a moderate amount of multiplication gain. The dark current and bandwidth were 1.3 μA and 27 GHz, respectively.

It should be mentioned that much of the recent development has been focused on CMOS-compatible processing techniques of Ge photodiodes [177, 178] including the thermal budget of Ge epitaxy and post-growth annealing [179]. To reduce the dislocation density it is often required to anneal the Ge at very high temperatures (≥900 °C). However, this anneal can degrade the CMOS-device performance if the Ge photodiodes layers are integrated after the formation of the CMOS electronics. In another approach Knoll et al. recently used a BiCMOS process to develop a waveguide Ge p–i–n photodiode on SOI waveguide [180]. The PD had a low dark current of 50 nA at 1 V and an internal responsivity of more than 0.6 A/W at 1.55 μm wavelength. The device showed a bandwidth of 35 GHz and was integrated with a high-performance SiGe:C HBT that was designed for 25 Gbit/s data detection (Fig. 6.25).

6.2.8 Heterogeneously Integrated Photodiodes

To integrate group III–V material based active devices onto SOI waveguides several hybrid and heterogeneous integration schemes have been developed. While these approaches are particularly interesting for transmitters on silicon [181], they also led to low dark current photodiodes with high efficiencies beyond 1.55 μm. Furthermore, heterogeneous integration has the potential to fully exploit bandgap engineering available in III–V materials to design more complex detector heterostructures. There have been five documented optical coupling schemes reported for III–V photodiodes on SOI waveguides (Fig. 6.26).

In [182], Park et al. demonstrated evanescently-coupled waveguide photodiodes utilizing an absorber consisting of both compressively and tensile strained AlGaInAs quantum wells wafer-bonded onto an SOI waveguide (Fig. 6.26(a)). The

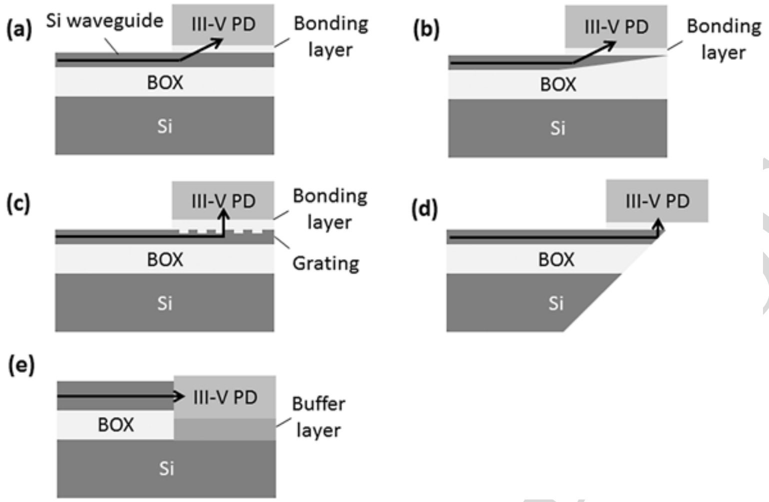


Fig. 6.26 Coupling schemes for III–V photodiodes on SOI waveguide: (**a**) evanescent, (**b**) adiabatic taper, (**c**) vertical coupling grating, (**d**) 45° polished facet or mirror, and (**e**) butt-coupling

photodiode layers were grown on an InP substrate and then transferred to the patterned silicon wafer using a low temperature oxygen assisted bonding process. The photodetector had low dark current of 100 nA at 2 V and fiber-coupled responsivities of 0.31 A/W and 0.23 A/W at 1.55 μm and 1.65 μm wavelengths, respectively. In [183] it was demonstrated that a similar bonding process can be used for selective area bonding to integrate two different III–V epitaxial layers on a silicon chip simultaneously. The demonstrated chip comprised an InGaAs/InP p–i–n photodiode and an AlInGaAs MQW laser that were part of a triplexer. Using a similar wafer bonding technology, Xie et al. recently demonstrated InP-based MUTC PDs on SOI waveguides for high-power high-speed applications [184]. The structure was adopted from a normal-incidence InGaAsP/InP MUTC photodiode that had previously achieved high saturation current and high linearity [185]. A schematic of the layer stack is shown in Fig. 6.26. The demonstrated bandwidth was 48 GHz and at 6.5 V bias the photodiode delivered more than 12 dBm RF output power at 40 GHz. Typical dark currents were below 10 nA at 5 V reverse bias voltage, and the internal responsivities were as high as 0.95 A/W at 1.55 μm wavelength. In [186] Piels et al. showed that InGaAs/InP p–i–n photodiodes can also be integrated on low-loss Si_3N_4 waveguides. The III–V material was wafer-bonded onto an intermediate silicon layer that, due to its lateral tapering, facilitated adiabatic coupling between the Si_3N_4 waveguide and the photodiode (Fig. 6.26(b)). A photodiode with $4 \times 30\ \mu m^2$ active area had a fiber-coupled responsivity of 0.36 A/W and 30 GHz bandwidth. Using a tapered InP membrane input waveguide to couple light out of a Si photonic wire on the SOI wafer, Binetti et al. demonstrated a p–i–n PD with a bandwidth of 33 GHz in [187]. Heterogeneous integration on SOI was achieved by direct molecular bonding of InP dies using a 300-nm-thick SiO_2 interface layer. For a photodiode with a mesa area of $50\ \mu m^2$ the responsivity and dark current were 0.45 A/W and 2 nA at

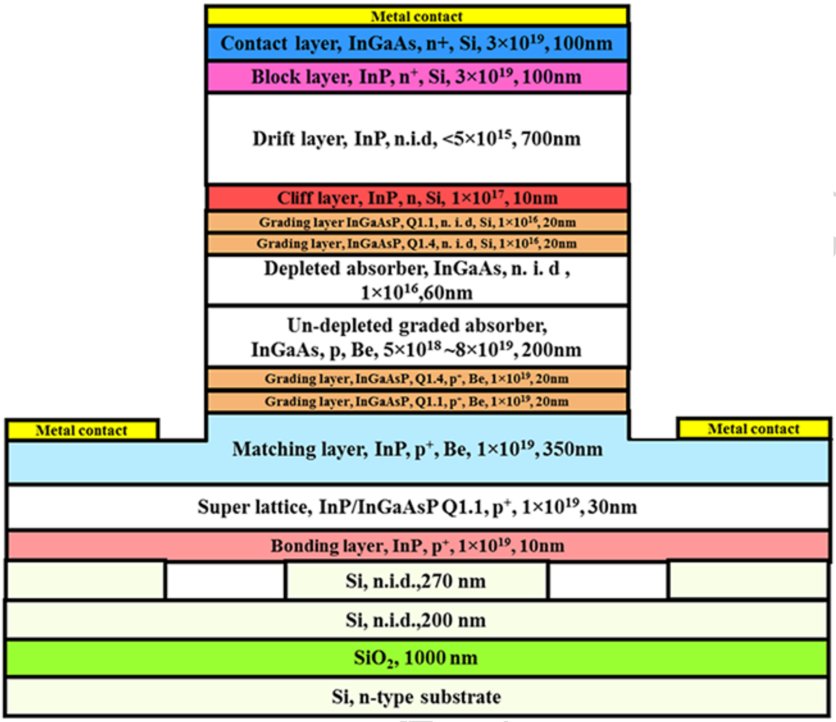


Fig. 6.27 Layer stack of heterogeneously integrated MUTC PD on SOI waveguide. Doping concentrations in cm^{-3}

4 V, respectively. In [188], Roelkens et al. demonstrated a grating coupler to diffract light from an SOI waveguide into an InP/InGaAsP photodiode (Fig. 6.26(c)). The InP die was bonded onto the SOI waveguide structure using adhesive bonding with benzocyclobutene (BCB) as a bonding layer. Since this vertical coupling scheme allows thicker bonding layers (∼3 μm), the requirements on roughness, flatness, and cleanliness of the two wafers that have to be joined, were somewhat relaxed. The measured dark current at 1 V and the responsivity at 1.55 μm were 0.3 nA and 0.02 A/W, respectively. The low responsivity was mainly attributed to the thin InGaAsP absorber that was only 120 nm. Recently, the same wafer bonding technology was also used to integrate GaInAsSb photodiodes onto SOI waveguides for detection in the SWIR range [189]. At 2.3 μm wavelength, photodiodes using evanescent coupling exhibited a responsivity of 1.4 A/W and devices utilizing a grating coupler achieved a responsivity of 0.4 A/W. The dark current was 4 μA at 1 V at room temperature.

In another approach shown in Fig. 6.26(d) the photodiode chip is integrated onto the silicon chip surface, where an etched or polished reflective mirror redirects the light beam to the detector surface [190]. Using a 45° polished facet, Zimmermann et al. demonstrated that high-speed vertically illuminated photodiodes can be assembled on the waveguide using epoxy [191]. Typical insertion loss and photodiode bandwidth were 5 dB and 28 GHz, respectively.

Geng et al. studied InGaAs pin photodiodes that were butt-coupled to SOI waveguides (Fig. 6.26(e)) in [192]. The authors used a selective-area MOCVD technique to grow the photodiodes on 2.6 μm-thick InP/GaAs buffers on Si. Photodiodes with an area of 64 μm^2 had a dark current of 400 μA and a bandwidth of 15 GHz at 5 V.

6.3 Summary

The continuing demand for higher and higher speed and the need for photonic integration have spurred the development of UTC-type structures and waveguide photodiodes. To date, high-efficiency evanescently coupled waveguide PDs with bandwidths well above 100 GHz have been demonstrated. Recently, there has been increased emphasis in achieving similar performance goals in photonic integrated circuits on Si. This has required novel column IV and III–V compound materials efforts, process developments, and optimized device designs. While significant progress has been achieved, challenges to reduce dark current and increase bandwidth-efficiency products in combination with achieving process compatibility with existing CMOS remain.

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